

JOOYOUNG YOO

CONTACT INFORMATION

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<https://jooyoung01.github.io>

EDUCATION

Emory University

Ph.D. in Computer Science (Advisor: Dr. Joon-Seok Kim)

PRESENT

University of Southern California

M.S. in Spatial Data Science (Advisor: Dr. John Wilson, and Dr. Yi Qi)

2025

Myongji University

M.S. in Data Technology (Advisor: Dr. Daewon Kim)

2020

Myongji University

B.E. in Information and Communication Engineering

2016

PUBLICATIONS

Jooyoung Yoo, Yi Qi, Isaac Ashe-McNalley, Beau MacDonald, and John P. Wilson (2026). “An Open and Transferable Deep Learning Framework for Mapping Urban Tree Canopy Using NAIP Imagery” *Remote Sensing* 18, no. 12: 1899.

Joon-Seok Kim, **Jooyoung Yoo**, Su Yeon Han, Andreas Züfle (2025), “An Agile Design of Activity-based Mobility Intervention using Large Language Models” *The 8th ACM SIGSPATIAL International Workshop on Geospatial Simulation (GeoSIM '25)*, Nov.3, 2025

Su Yeon Han, Yubin Lee, **Jooyoung Yoo**, Jeon-Young Kang, Jinwoo Park, Soe W. Myint, Eunsang Cho, Xin Gu, and Joon-Seok Kim (2025). “Spatial disparities in fire shelter accessibility: Capacity challenges in the palisades and eaton fires.” arXiv preprint arXiv:2506.06803.

Su Yeon Han, **Jooyoung Yoo**, Alexander Michels, Shaowen Wang, Joon-Seok Kim (2025), “Vulnerable Neighborhood Explorer (VNE): An Open-Source Visual Analytics Tool for Exploring Social Vulnerability to Disasters across Different Neighborhoods” *SoftwareX*, 31, p.102233.

Jingu Kim, Eun Young Kang, **Jooyoung Yoo** (2025), “Exploring Social Interdependence and Peer Relationships of University Students: A Longitudinal Social Network Analysis.” *Journal of Educational Innovation Research*, 35, no. 2: 25–46.

Siqin Wang, **Jooyoung Yoo**, Wenhui Cai, Fan Yang, Xiao Huang, Qian Chayn Sun, Shaokun Lyu (2025). “Reducing the social inequity of neighborhood visual environment in Los Angeles through computer vision and multi-model machine learning.” *Sustainable Cities and Society*, 119, p.106062.

Jooyoung Yoo, Reem Emad Shtaiwi, Mohammad Yasin, Dweep Trivedi, Abdullah Alfarrarjeh, Amani Abu Jabal, Seon Ho Kim (2024), “Towards real-world deployment of deep learning solutions for global road damage detection and classification,” *IEEE Big Data, Big Data Cup Challenge (ORDDC'2024)*, **awarded the Bronze Prize in an international challenge**, pp. 8485-8494. Dec.17, 2024

Jooyoung Yoo, Abdullah Alfarrarjeh, Krish Sukhani, Seon Ho Kim, and Cyrus Shahabi (2024). “STVD: Synchronized Truck Video Dataset with RGB and IR cameras for continuous truck traffic monitoring,” *IEEE International Workshop on AI and Computer Science*, Dec.9, 2024

Jooyoung Yoo, Seon Ho Kim, Krish Sukhani, Min Sang Yoo, and Cyrus Shahabi (2024). “Truck detection and counting in low-light condition: do we need infrared camera?,” *In IEEE International Conference on Big Data and Smart Computing*, Feb.15, 2024

Jooyoung Yoo, and Daewon Kim (2020). “Development of a new pedestrian avoidance algorithm considering a social distance for social robots,” *Journal of Broadcast Engineering* 25.5 : 734-741.

Jooyoung Yoo, and Jin Lee Park (2020). “Examining the socialization of grit and the effects of peer community and teacher closeness using longitudinal social network analysis,” *Journal of Learner-*

Jooyoung Yoo, Joomin Kim, Sungsik Yun, and Daewon Kim (2013). “Development of a scenario-based work distribution function for teleoperation under multi-user and multi-robot environments,” *IEEE International Symposium on Robotics 2013*

RESEARCH
EXPERIENCE

Urban World Model and Multimodal Spatial Representation Project JUL. 2026 – PRESENT
Ph.D. Student (Advisor: Dr. Joon-seok Kim)

- Developing an urban world model that learns unified spatial representations from heterogeneous geospatial data, including aerial imagery, LiDAR, OpenStreetMap, and urban attributes.
- Investigating multimodal embedding and fusion methods to align different representations of the same urban environment into a shared latent space.
- Exploring generative world models and 3D simulation environments for representing and simulating real-world urban environments.

Individual Tree Detection, Segmentation for Urban Greening Project MAY. 2024 – AUG. 2025
Full-time Student Researcher (Advisor: Dr. John Wilson, and Dr. Yi Qi)

- Developed detection and segmentation models using aerial imagery to identify and segment individual trees.
- Designed a training and prediction workflow for LA city administrators, enabling them to count trees and calculate canopy areas using GIS, without the need for coding skills.

Spatial Pattern Mapping using Street View: Human Perception Project MAR. 2024 – SEP. 2024
Volunteer Student Researcher (Advisor: Dr. Siqin Wang)

- Conducted data collection at 50-meter intervals across Los Angeles County using GIS software, Python libraries, and 360-degree imagery from Google and Mapillary Street View APIs.
- Applied machine learning methods, including segmentation and classification, to analyze Street View imagery for sentiment analysis of various locations.
- Visualized spatial patterns of six human emotions (Beautiful, Wealthy, Livable, Safe, Boring, Depressing) to map urban perception.

EMPLOYMENT

Spatial Data Lab (SDL) at Harvard University, *Intern* 2024 - 2025

Spatial Sciences Institute of USC, *Database Analyst, Full-time Researcher* 2024 - 2025

Keck School of Medicine of USC, *Full-time Student Researcher* 2023 – 2024

Integrated Media Systems Center of USC, *Teaching Assistant, Student Researcher* 2023 – 2024

Joongbu University, *Web Developer* 2020 – 2022

Techsphere, *Computer Vision AI Software Engineer* 2020 – 2021

Myongji University, *Research Assistant, Teaching Assistant* 2018 – 2020

Konkuk University, *Web Developer* 2018

Korean Dictionary, *Co-founder* 2013 – 2017

[†]Non-profit Organization

AWARDS

Winner – 2026 Esri Innovation Program Students of the Year, Esri 2026

Bronze Prize – Optimized Road Damage Detection Challenge 2024, IEEE BigData 2024

Best Presenter – The 23rd KOCSEA Technical Symposium 2023

Best Presenter – USC CKIDS DataFest Fall 2022 2022

SKILLS

DL Frameworks: Pytorch, Keras, Tensorflow, Ultralytics

Languages: Python, R, C++, Java, JavaScript, SQL

Other: Git, Linux, Matlab, ArcGIS Pro, QGIS, SPSS, L^AT_EX – All professional proficiency or above